

Kaung Nyo Lwin

AI ENGINEER & RESEARCHER



Computer Vision

LLM Fine-Tuning & RL

Biometrics & Fairness

MLOps on GCP

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PROFILE

AI engineer and researcher who takes vision and language models from experiment to production on Google Cloud, backed by rigorous, standards-based evaluation. At AIT I build face-liveness benchmarks for digital identity (MOSIP), calibrated camera measurement for driving tests and video conflict analytics for road safety. My thesis, L2BC, trains compact LLMs with GRPO on rule-verified rewards, letting a 3B model beat 7B agentic pipelines.

EXPERIENCE

Computer Vision Researcher

Asian Institute of Technology · MOSIP (Modular Open Source Identity Platform)

Jul 2026 – Present

Face liveness and presentation-attack detection (PAD) for the Inji digital-identity wallet, with an APAC fairness focus

- Built a PyTorch PAD research platform (5 benchmark loaders, a 45-model open-source survey, TFLite, ONNX and Torch adapters) reporting ISO/IEC 30107-3 metrics (APCER, BPCER, ACER, bootstrap CIs) with per-ethnicity fairness.
- Benchmarked the Inji wallet's liveness SDK models on four PAD benchmarks (247,840 frames), uncovering inverted score polarity and mismatched tensor layouts across released TFLite assets; added contract probes.
- Ran 38+ controlled training runs (MobileNet, ResNet, EfficientNet), including a physically based relighting ablation; a four-dataset mixture cut cross-dataset ACER by 73% yet widened ethnic disparity from 1.6× to 6.7×.
- Validated ISO/IEC 29794-5 (OFIQ) occlusion measures on eyewear labels (87.8% accuracy, AUC 0.90); designing a VLM-assisted occlusion detector with Grounding DINO, SAM 2 and Qwen2.5-VL.

Computer Vision Engineer

Asian Institute of Technology · CDG Group

Jul 2026 – Present

Driver Licensing Management System (DLMS): automated driving-test assessment from pole-mounted cameras

- Designed and built a human-in-the-loop vision pipeline (RTSP capture, calibration, marker tracking, rule checks, scoring) that turns test-station video into metric vehicle trajectories for rule-based scoring, backed by 450+ tests.
- Engineered the camera geometry (homography and PnP solvers, a two-plane ground/roof-marker model, lens fitting from straight lines), reaching 1.14 mm mean (2.59 mm p95) error within 15 m on exact synthetic ground truth.
- Built a fault-tolerant RTSP recorder (lossless stream copy, stall detection, auto-reconnect), camera focus control over HTTP, and a browser-based studio for control points, regions of interest and scoring lines.

Research Engineer

Asian Institute of Technology · Toyota Mobility Foundation (TRUST project)

Mar 2026 – Present

Road-safety research platform combining vehicle telemetry and roadside CCTV in Bangkok

- Built the video conflict-analysis pipeline: YOLO and ByteTrack tracking, homography-based ground-plane trajectories, kinematics and surrogate safety measures (TTC, PET, DRAC), BigQuery outputs and 2D map replays.
- Trained a 5-class YOLO11-L road-user detector on annotated project CCTV (mAP50 0.76, mAP50-95 0.55), with containerised training and batch inference on Vertex AI and Cloud Run, GCS checkpoints and a model registry.
- Derived depth-zoned measurement windows from a noise analysis of 3.1M detections, extending reliable speed estimates from 38 m to 70 m and acceleration events from 21 m to 54 m.
- Built Poisson and negative-binomial GLM tests (exposure offsets, rate ratios, boundary sensitivity) of a warning zone's effect on harsh braking over 700K telemetry pings from 21K+ vehicles.
- Migrated the platform to Google Cloud: FastAPI on Cloud Run, a layered BigQuery analytics module, and Terraform-managed Cloud Build, Secret Manager and storage behind the research dashboard's insight APIs.

LLM Developer Intern

AI Brain Lab

Jul 2025 – Oct 2025

Generative AI for ESG reporting and financial-document question answering

- Built an ESG report-generation pipeline with LLMs and retrieval-augmented generation (RAG), and a FastAPI and OpenAI API assistant answering questions over Stock Exchange of Thailand (SET) documents.
- Scraped and embedded thousands of SET documents with RAPTOR tree-structured retrieval, and implemented table extraction and structured parsing for PDF, Excel and CSV sources.

● RESEARCH & PUBLICATIONS

Research interests Verifiable rewards and RL for LLMs; fair, trustworthy biometrics; geometric computer vision

L2BC: Reinforcement Fine-Tuning for Text-to-Visualization

Master's thesis · Asian Institute of Technology

2026

L2BC ("Learn To Be Cool"): small code LLMs learn to chart data from rewards that are verified, not judged

- Proposed, to our knowledge, the first reinforcement fine-tuning framework for text-to-visualization with a purely rule-based reward: programs are executed, and their rendered charts parsed and verified against ground truth.
- Turned the verified reward's expectation ("comfortability") into a training-state signal that sets GRPO's temperature, learning rate and reward shape each epoch, with no LLM or VLM judge in the loop.
- Fine-tuned Qwen2.5-Coder 1.5B and 3B with LoRA and TRL on 192 VisEval queries; the 3B model reaches 67.97% single-table and 59.26% multi-table pass rates in one pass, beating 7B agentic baselines (59.87%, 43.32%).
- Reward decomposition and ablations (workflow structure, scale, chain-of-thought) show what verification teaches and where it stops: unrewarded sort-order errors grow on unseen tasks.

K. N. Lwin, T. Tangtortan, C. Silpasuwanchai. *L2BC: Reinforcement Fine-Tuning for Text-to-Visualization with Rule-Based Rewards over Parsed Chart Schemas*. Manuscript in preparation, 2026.

● EDUCATION

○ Master of Engineering, Data Science and Artificial Intelligence

Asian Institute of Technology, Pathum Thani, Thailand

2024 – 2026

○ Bachelor of Engineering, Avionics

Myanmar Aerospace Engineering University, Meiktila, Myanmar

2010 – 2015

● TECHNICAL SKILLS

Machine learning	PyTorch, Ultralytics YOLO, ONNX, TensorFlow Lite, scikit-learn, SciPy, NumPy, pandas
LLMs & generative AI	GRPO, PPO, DPO, LoRA, Hugging Face Transformers, TRL, RAG, LlamaIndex, LangChain, OpenAI API, FAISS, pgvector, agents and MCP
Computer vision	Object detection, multi-object tracking (ByteTrack), OpenCV, camera calibration (homography, PnP, lens distortion), face anti-spoofing, OFIQ
Evaluation & statistics	ISO/IEC 30107-3, fairness (ISO/IEC 19795-10), bootstrap CIs, GLMs (statsmodels), ablations
MLOps & cloud	Google Cloud (Vertex AI, Cloud Run, Cloud Build, BigQuery, Cloud Storage), Terraform, Docker, FastAPI, MLflow, pytest, Git
Data & languages	Python, SQL, PostgreSQL, MySQL, SQL Server, MongoDB, Shapely, Power BI, Odoo, PowerShell

● SELECTED PROJECTS

- **Software Documentation Bot.** LLM pipeline that drafts software documentation for business users, reducing manual hand-offs between developers and stakeholders (Python, LlamaIndex, Flask).
- **Job Automation Risk Prediction.** Machine-learning model scoring each occupation's automation risk, served in an interactive dashboard with career recommendations (scikit-learn, Django).

● ADDITIONAL EXPERIENCE

○ Solutions Architect (Contract) · MA'HAR EMPEROR (Audit Firm)

Aug 2026 – Present

Own end-to-end delivery of a project management and tracking system on Odoo 18 and PostgreSQL: service-hour planning, timesheets, rate and margin resolution, role-based access and management reporting.

○ Senior Business Analyst · KEYfields

Feb 2023 – Jul 2024

Engineered SQL stored procedures and data models for SaaS accounting and logistics products; automated deployments and database backups with Python and PowerShell, cutting processing errors and runtime by 50%.

○ Asset Management Analyst, promoted to Senior · Telecom International Myanmar

Jun 2017 – Sep 2020

Built Excel VBA automation and KPI dashboards for 100+ asset types, reducing manual reporting by 40%; coordinated rollouts across 700+ telecom sites at 98% KPI compliance.

○ Telecom Installation Engineer · Ericsson Myanmar

Feb 2016 – May 2016

● CERTIFICATIONS & LANGUAGES

Certifications DeepLearning.AI: Deep Learning and Machine Learning Specializations, TensorFlow Developer, Mathematics for Machine Learning and Data Science. IBM: Data Science, Data Analyst.

Languages Myanmar (native), English (C1)